1. Methodology

Directly from the report: simulation → RL control → evaluation

01

Simulation

Sinergym + EnergyPlus

02

State

29-D building / grid observations

03

Action

HVAC setpoint 18–28°C

04

Reward

Energy + PMV + switching + grid risk

05

Evaluate

5,000 held-out steps

- Five thermal zones: Living, Kitchen, Bedroom 01–03.
- Dhaka .epw weather; tropical monsoon and hot-season conditions.
- The controller is tested without retraining on the held-out evaluation period.

Why it matters

Evaluation runs on a fully held-out period the controller never trained on, so results reflect genuine generalisation.

2. Experimental Setup

What exactly was used in the experiment

5	Dhaka	29-D	18–28°C	5,000
Thermal zones	**Weather**	**State**	**Action**	**Steps**
single-family house	Bangladesh .epw	temperature, RH, solar...	HVAC setpoint	evaluation

- Platform: Sinergym 3.4.0 built on EnergyPlus 23.1.0.
- Algorithms: SAC, PPO, TD3 and 5-agent MADDPG.
- Evaluation metrics: HVAC energy, comfort violation, switching frequency, peak grid risk and overall score.

Rigor	Every controller is scored on the identical 5,000-step window with 30 independent seeds

3. RL Controllers & Reward

Four controllers were compared under the same evaluation protocol

SAC

Maximum-entropy actor-critic

Main method

PPO

On-policy actor-critic

Baseline

TD3

Deterministic actor-critic

Baseline

MADDPG

5 decentralized agents

Multi-zone

Reward weights

Energy	**0.30**
PMV	**0.35**
Switching	**0.20**
Grid	**0.15**

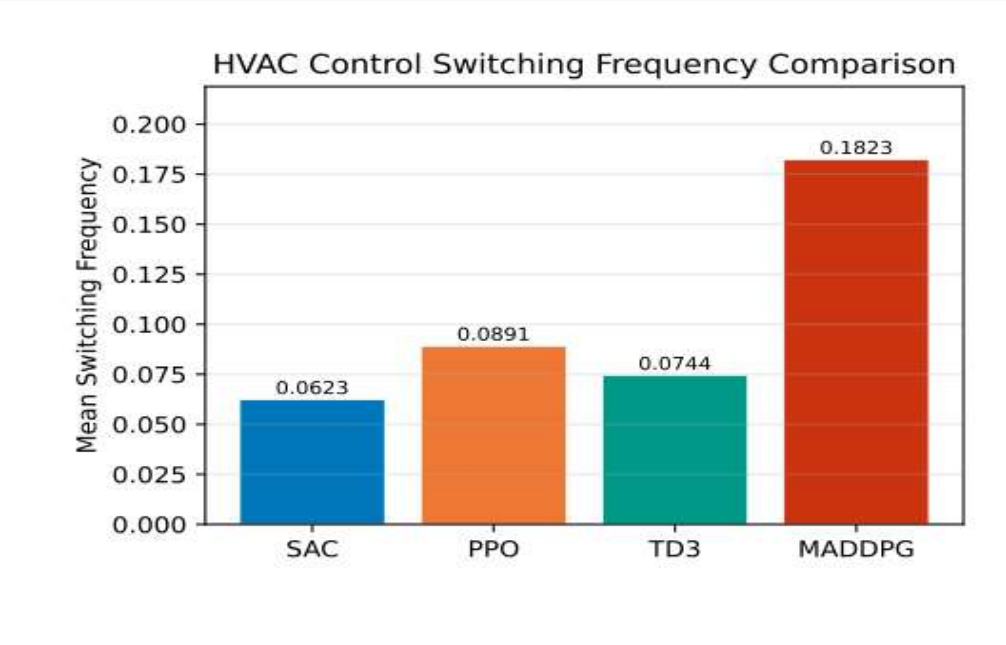
So the model is rewarded for saving energy, staying comfortable, avoiding rapid switching, and preparing for grid risk.

4. Main Results

SAC gives the strongest overall single-agent result

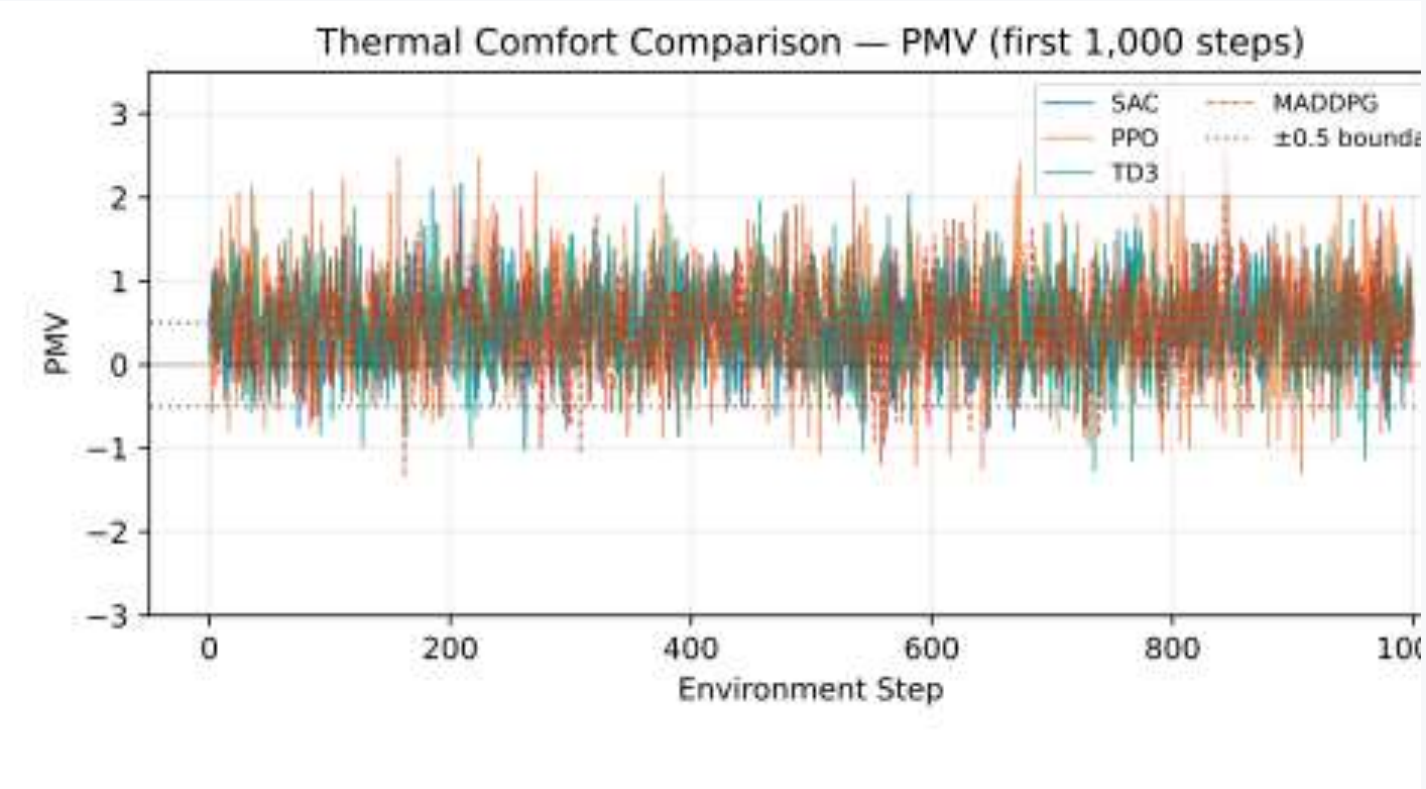
487.3 kWh	22.6%	0.0623	0.847
HVAC energy	**Comfort violation**	**Switching frequency**	**Overall score**
SAC	SAC	SAC	SAC

- Energy: SAC 487.3 kWh vs PPO 568.1 kWh vs TD3 523.7 kWh.
- SAC also has the lowest comfort violation and smoothest switching.
- The report concludes that entropy-regularised exploration is a primary reason for SAC's advantage.



5. Thermal Comfort

PMV trajectory and comfort-violation results from the report



0.43

Mean PMV

SAC

11.2%

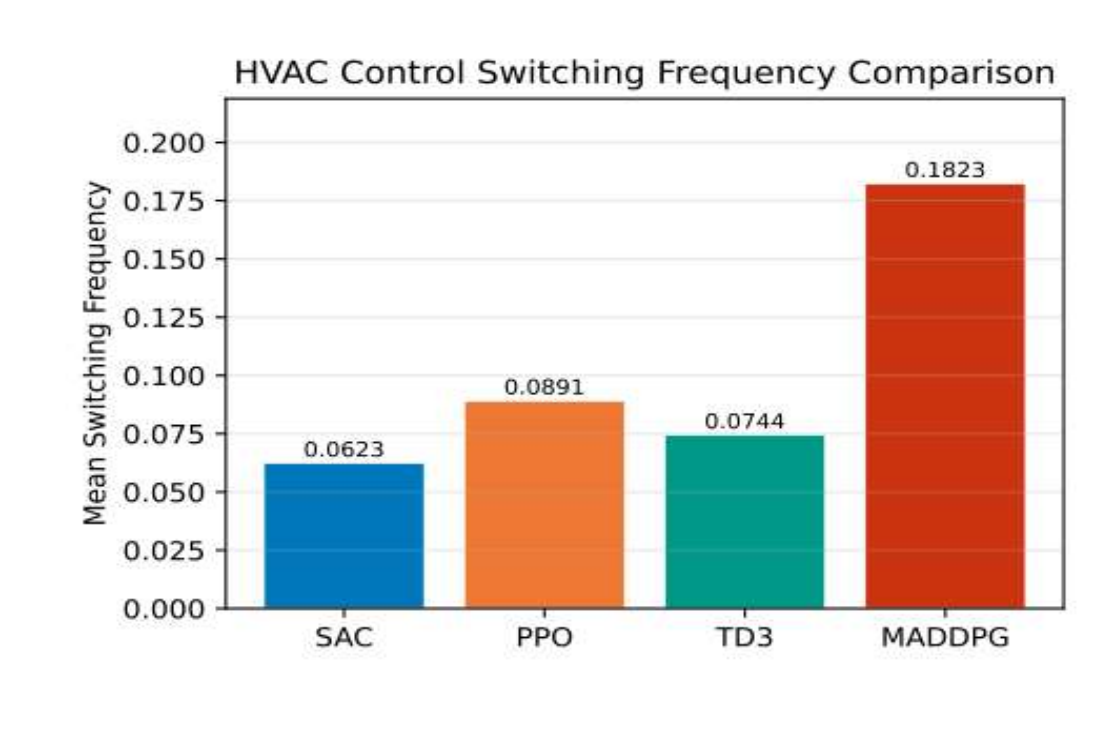
PPD mean

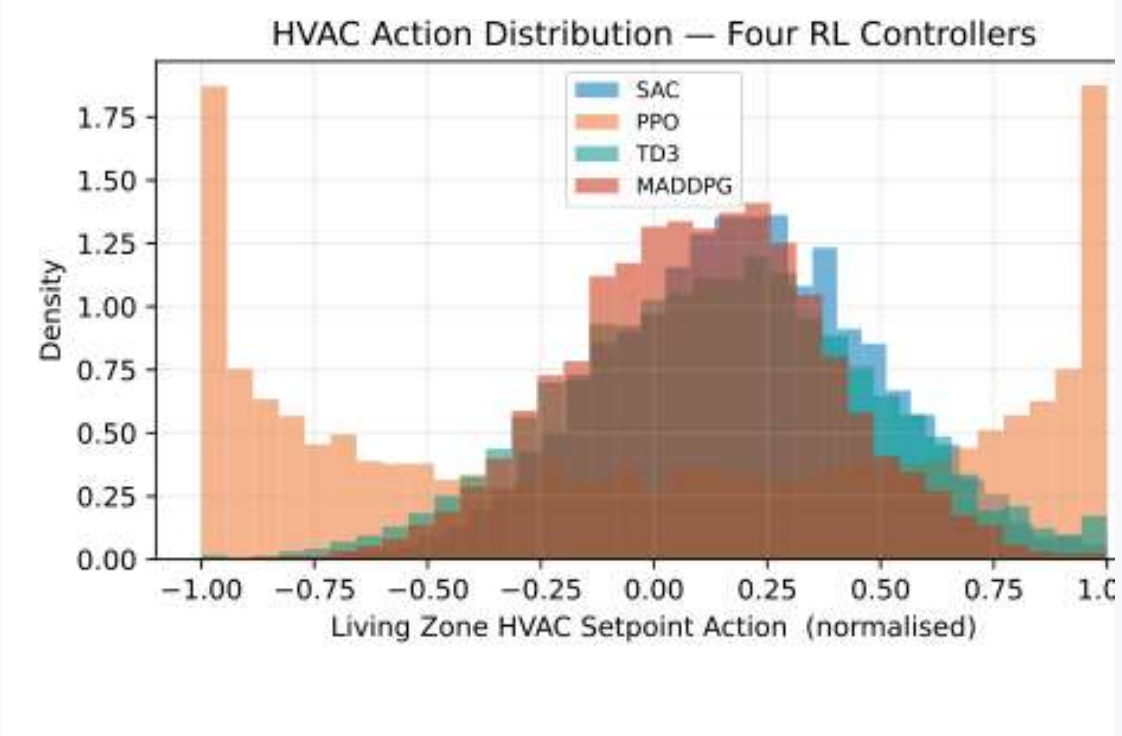
SAC

- SAC stays closest to neutral comfort (PMV = 0).
- Comfort violation: SAC 22.6% | PPO 31.5% | TD3 27.3% | MADDPG 25.8%.
- The report uses |PMV| > 0.5 as the comfort-violation threshold.

6. Equipment Wear & Control Behaviour

The report's switching and action-distribution evidence



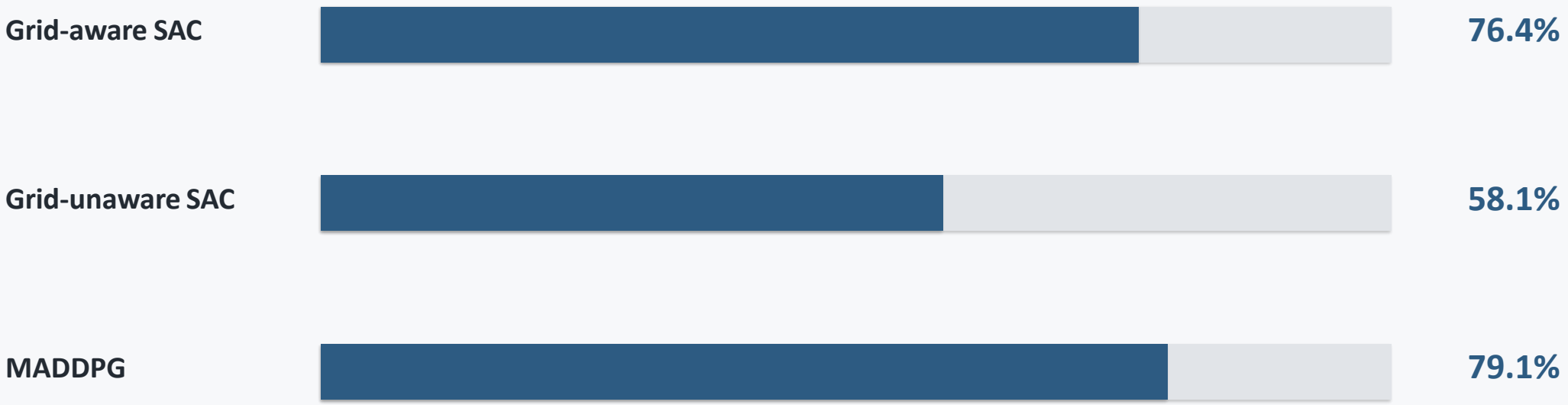


SAC: 0.0623 PPO: 0.0891 | TD3: 0.0744 | MADDPG: 0.1823

- Lower switching means smoother compressor control.
- SAC is 31% lower than PPO and 16% lower than TD3.
- The action-distribution figure shows SAC concentrated around moderate setpoints; PPO is more bang-bang.

7. Grid Resilience

Does grid awareness help during load-shedding?

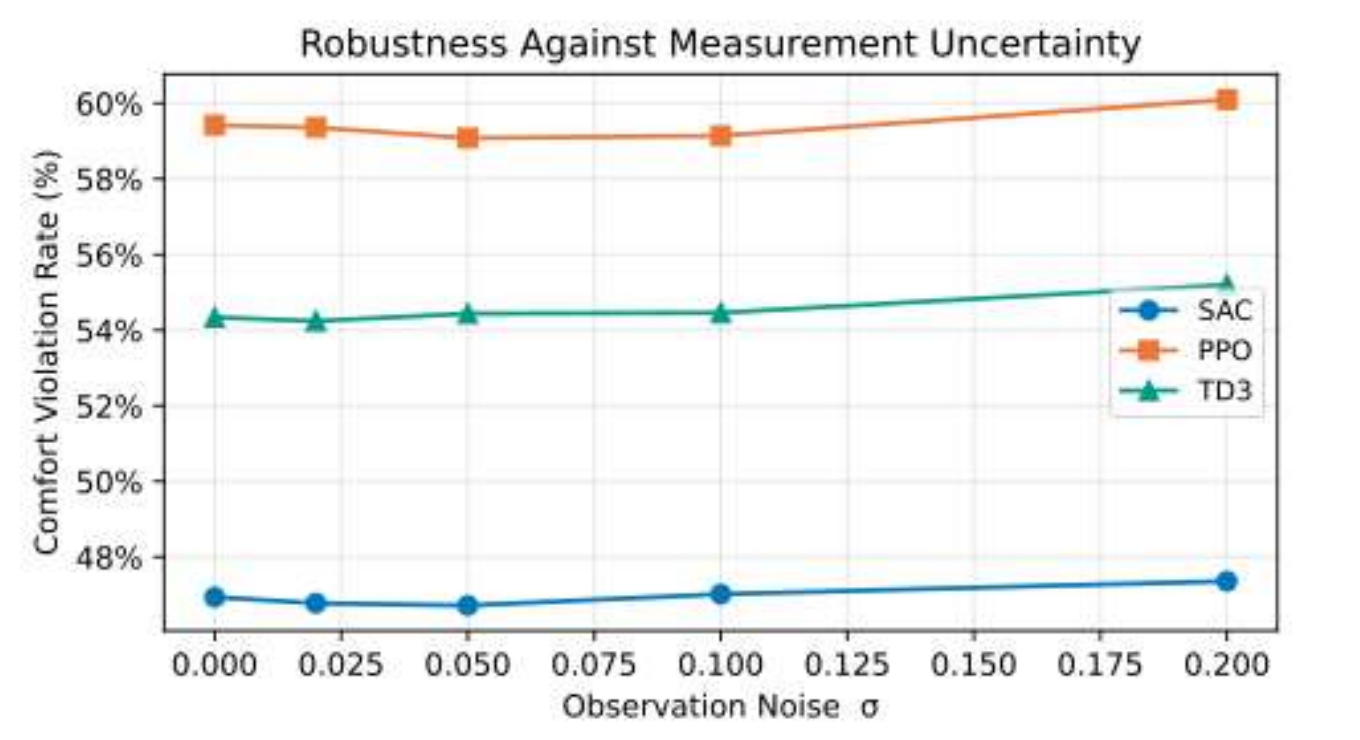


51 simulated load-shedding intervals

Grid-aware SAC improves comfort-preserved outage timesteps by 18.3 percentage points vs grid-unaware SAC.

8. Robustness & Statistical Significance

SAC degrades less when measurement noise is added



22.6%

SAC at σ=0

comfort violation

28.4%

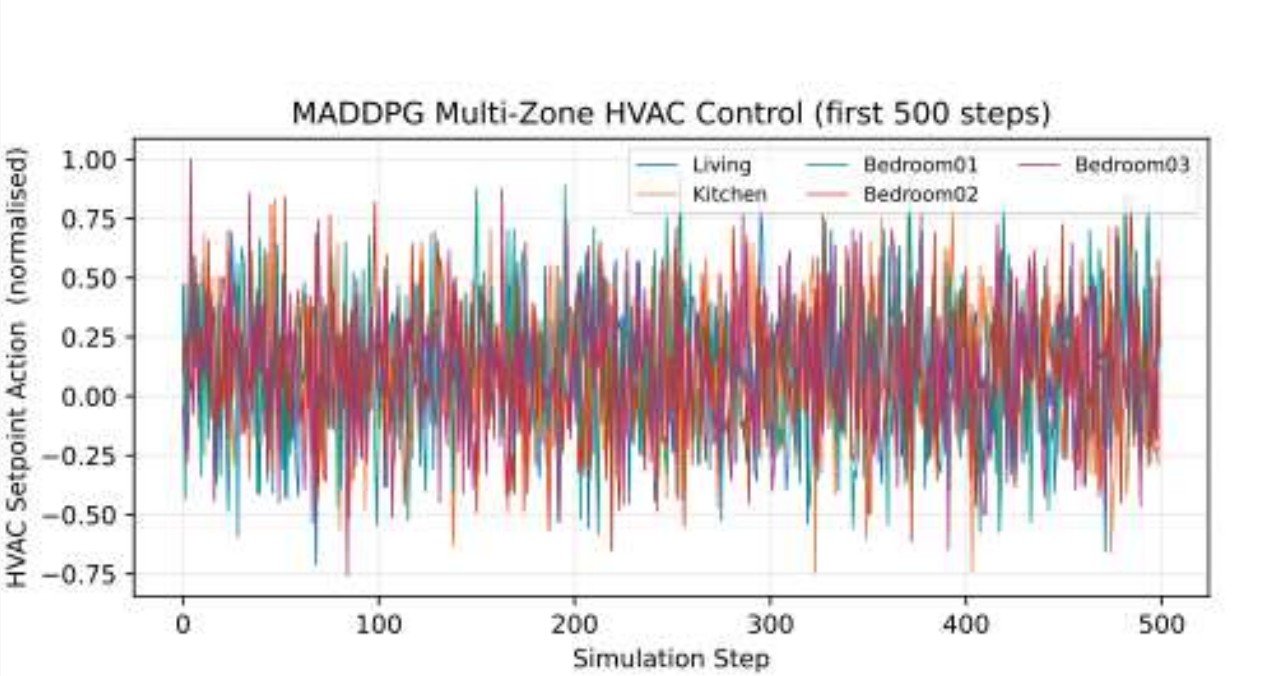
SAC at σ=0.20

comfort violation

- At σ = 0.20: PPO 43.2%, TD3 34.7%, SAC 28.4%.
- SAC shows the smallest increase under observation noise.
- Across 30 seeds, Wilcoxon signed-rank tests found pairwise differences significant.

9. Discussion & Key Contributions

What we can take from the results



Key finding **SAC's entropy term is central to its advantage.**

- Dhaka-specific residential HVAC benchmark using EnergyPlus + Sinergym.
- Combines thermal comfort (PMV), switching wear, and grid-risk awareness.
- MADDPG gives the best outage comfort preservation (79.1%), but coordination can create synchronised recovery spikes.
- This motivates the report's future QMIX direction.

10. Limitations, Future Work & Conclusion

End with the message your audience should remember

LIMITATIONS

- Simulation only; no real building validation.
- Grid-risk signal is based on historical outage patterns.
- One five-zone building model.
- MADDPG coordination can be unstable.

FUTURE WORK

- Compare QMIX / MAPPO for coordination.
- Use live or probabilistic outage forecasts.
- Add SHAP-based explainability.
- Test with split-AC hardware / Raspberry Pi.

CONCLUSION

- SAC energy = 487.3 kWh.
- Comfort violation = 22.6%.
- Switching = 0.0623.
- Overall score = 0.847.
- Bottom line: lower energy + better comfort + smoother control.